

Marla Eisman

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EDUCATION

University of California, Berkeley

PhD, Mechanical Engineering (Controls) – GPA: 3.7/4.0

Berkeley, CA
Aug 2025 – May 2030 (expected)

University of Florida

B.S., Mechanical Engineering – Upper Division GPA: 3.85/4.0

Gainesville, FL
Aug 2020 – Dec 2024

PUBLICATIONS

M. Eisman, B. Lam, S. Sonnino, and F. Borrelli, “Racing a Wheeled Quadraped: Active Load Transfer Mitigation via Model Predictive Control,” *17th International Symposium on Advanced Vehicle Control*, 2026.

C.F. Nino, O. Sudhir Patil, J.C. Insinger, **M.R. Eisman**, and W.E. Dixon, “Online ResNet-Based Adaptive Control for Nonlinear Target Tracking,” *IEEE Control Systems Letters* / IEEE Conference on Decision and Control, 2025.

RESEARCH EXPERIENCE

Model Predictive Control Lab, UC Berkeley (Advisor: Prof. Francesco Borrelli)

Graduate Researcher

Berkeley, CA
Aug 2025 – Present

- Developed MPC for racing a wheeled quadraped with active load transfer mitigation (AVEC 2026).
- Integrated a Unitree Go2W wheeled quadraped into a modular ROS 2 interface, fusing motion-capture data with on-board state estimation to create an open-source, lab-wide platform for controller evaluation.
- Researching MPC and RL for energy management, raceline optimization, and stability through obstacles.

Nonlinear Controls and Robotics Lab, University of Florida (Advisor: Prof. Warren Dixon)

Undergraduate Researcher

Gainesville, FL
Jan 2023 – Mar 2025

- Designed a deep residual neural network architecture with a Lyapunov-stable adaptive controller for nonlinear target tracking; validated theoretical predictions in simulation (IEEE L-CSS / CDC 2025).
- Built custom IP55-rated quadcopters and a ROS 2 framework streaming GPS, IMU, and lidar data over a 5G Starlink network; demonstrated multi-agent stability under communication interruptions to the US Air Force.
- Applied adaptive control to multi-agent herding and swarms; built a Python framework for multi-agent adaptive control and deep learning simulation.

INDUSTRY EXPERIENCE

Vatn Systems

Autonomy Engineering Intern

Bristol, RI
Mar – Aug 2025

- Designed and implemented a model-based LQR controller coupling 6-DOF AUV dynamics to fins, rudder, and thruster; validated in Python/C simulation and real-world underwater testing.
- Contributed to the company’s first multi-AUV swarm demo, integrating acoustic comms, Doppler velocity logging, and machine learning.

SpaceX

Final Integration Engineering Intern

Cape Canaveral, FL
May – Aug 2023

- Invented Dragon capsule seat protection hardware via FEA and material testing, saving \$7,900 per mission.
- Formulated hazardous propellant handling standards with automated valve leak detection, cutting 35 engineering tickets per mission.

Tesla

Thermal Management Engineering Intern

Fremont, CA
May – Aug 2022

- Built seven electromechanical test fixtures and a hardware-in-the-loop Python program for automated data logging and fault detection over 1000+ hour thermal cycles (−50 to +60 °C), cutting the testing timeline by nine months.

Gator Motorsports (Formula SAE Electric)

Lead Engineer, Vice President

Gainesville, FL
Aug 2020 – Apr 2023

- Led development of UF’s first electric Formula SAE vehicle, including a 400 V lithium-ion powertrain with liquid and air cooling and a MATLAB lap dynamics model with regenerative braking; raised \$70,000+ in funding.

SKILLS

Control & Robotics: Model predictive control, nonlinear/adaptive control, Lyapunov analysis, reinforcement learning, state estimation, multi-agent systems

Software & Hardware: Python, C++, MATLAB, ROS 2, FEA, SolidWorks

AWARDS & SERVICE

UC Berkeley First-Year Fellowship (2025) · Blue Origin Design Grant (2023) · Formula SAE Michigan 2022: 3rd Overall, 1st Acceleration (100 teams) · Teaching Assistant, UF MAE (2024) · SWE STEM outreach (2020–2024)